



UNICITY PROTOCOL • SEED PROPOSAL • 2026

# SECURING THE AGENTIC ECONOMY.

Proof of identity and permission, bound to the payment itself.

---

MIKE GAULT • CEO • [mike@unicity-labs.com](mailto:mike@unicity-labs.com)  
ALAN RADI • COMMERCIAL • [alan@unicity-labs.com](mailto:alan@unicity-labs.com)

PREPARED FOR  
GREG KIDD @ HARD YAKA

# THE MACHINES HAVE STARTED **PAYING EACH OTHER.**

Most of the internet is already machines – and they transact at software speed.

# 57.5%

of web requests are automated,  
not human – Cloudflare, 2026

About **100M on-chain payments** crossed Base in nine months, and Mastercard now runs an agent-payment system for machines. The amounts are tiny; the old payment systems cannot carry them. **The proof has to travel with the money.**

# FIRST WE CONVERT STABLECOINS TO OFF-CHAIN BEARER INSTRUMENTS.

## PLASMA USDT



an entry on a shared ledger

OFF-CHAIN



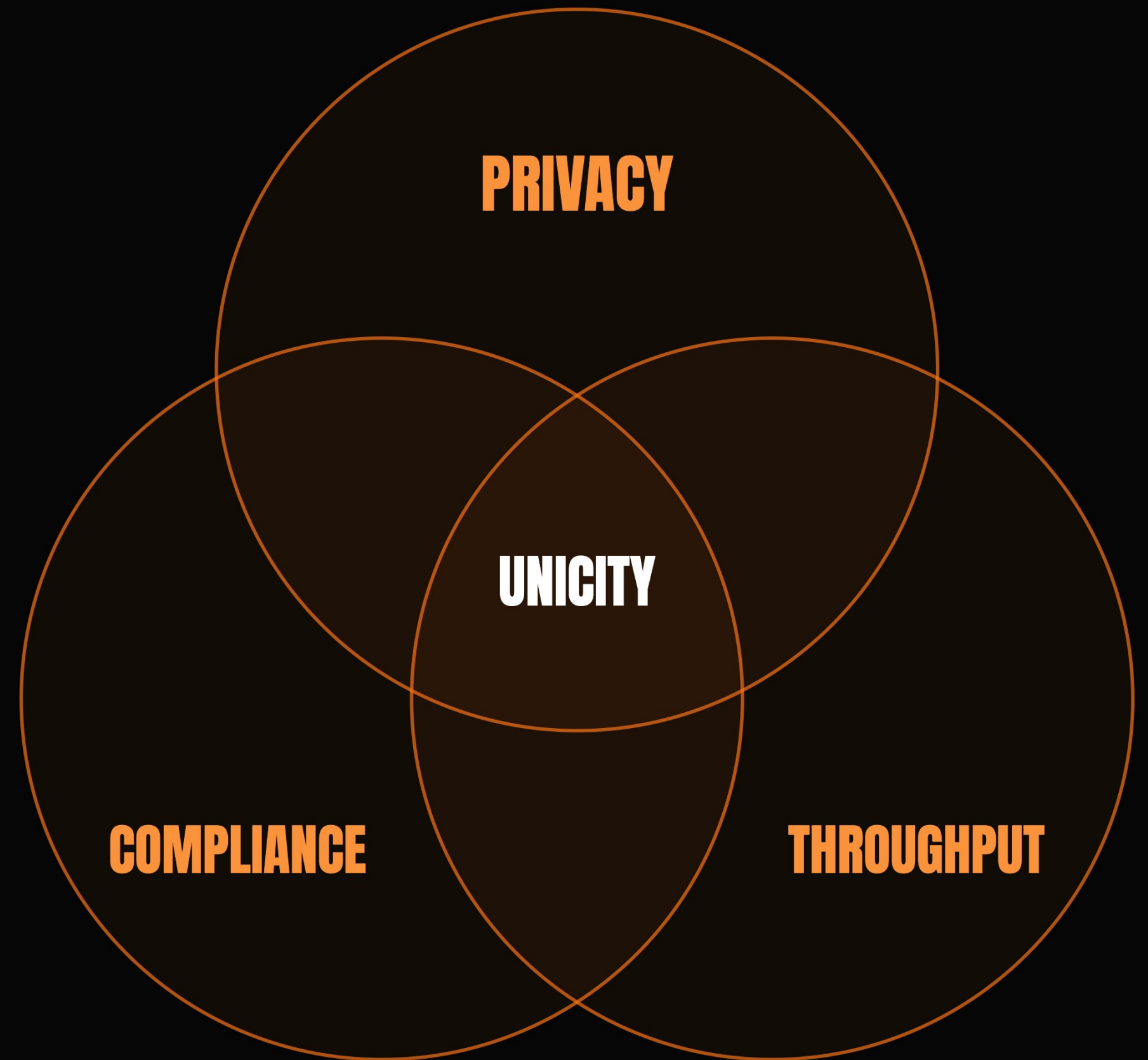
**BEARER INSTRUMENT**

- MOVES P2P
- PRIVATE
- COMPLIANT

# THE TRILEMMA

On a shared ledger every transaction is public – easy to audit, impossible to keep private. Add cryptography (ZK, FHE) to recover privacy and throughput collapses.

Unicity solves this trilemma by introducing bearer instruments.





# FAIR ACCESS TAKES THREE THINGS. **THE THIRD WAS NEVER BUILT.**

We solved disintermediation and digital value. Counterparty verification remains broken.

- |           |                                                                                                               |          |
|-----------|---------------------------------------------------------------------------------------------------------------|----------|
| <b>01</b> | <b>CUT OUT THE MIDDLEMAN</b><br>A decade of open infrastructure settled it.                                   | SOLVED   |
| <b>02</b> | <b>LET VALUE MOVE FREELY</b><br>Stablecoins now settle tens of trillions a year.                              | SOLVED   |
| <b>03</b> | <b>PROVE WHO IS ON THE OTHER END</b><br>No blockchain can verify a recipient before the transaction executes. | UNSOLVED |

Unicity was engineered to close **the third gap.**

# FROM SIDE-CAR TO **PROTOCOL**

Identity must live inside the asset, not beside the transaction.

## THE SIDE-CAR

Self-sovereign identity stalled because the credential sat beside the transaction. A check that can be skipped gets skipped.

## THE PROTOCOL

USBC drew the right conclusion: put identity inside the dollar. Unicity makes that embedded identity enforceable across any jurisdiction.

# TOKENS AS BEARER OBJECTS

an asset you **hold**, not a ledger entry the network maintains

## SELF-CONTAINED

value lives in the object

## SELF-PROVING

carries its own proof

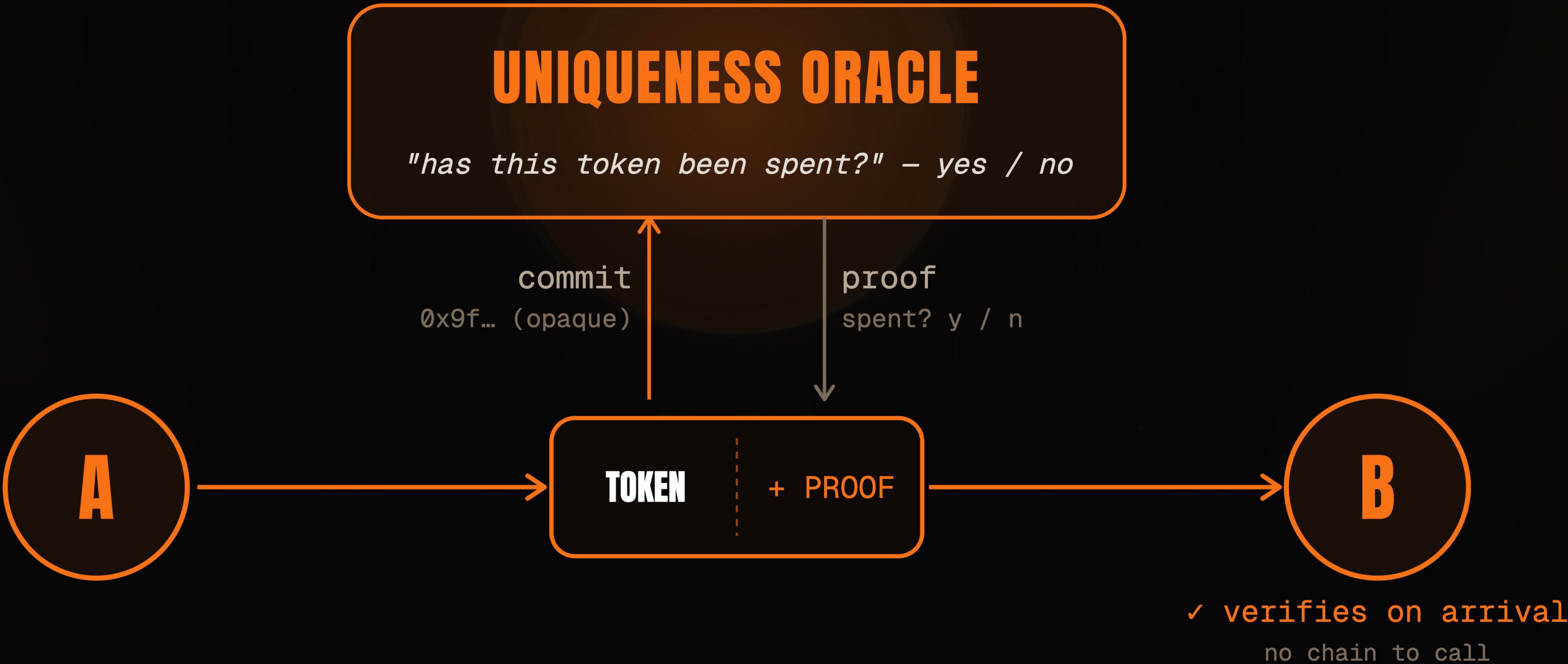
## PEER-TO-PEER

moves with no ledger lookup



# UNICITY GENERATES PROOFS OF UNIQUENESS (SINGLE SPEND)

Our invention is a **Uniqueness Oracle** – decentralized infrastructure that **proves the absence of double-spend without seeing transactions.**





# UNICITY PROVES UNIQUENESS: **NOTHING ELSE**

Proving no double-spend is the Oracle's only job. Transaction validation, storage and P2P communication are all **off-chain** – handled with traditional tools, by the parties who care.

OFF-CHAIN · TRADITIONAL TOOLS

## COMMUNICATION

the asset carries its own proof  
– the wire just moves bytes

NOSTR

EMAIL

SOCIAL

HOLEPUNCH

## STORAGE

state lives with the owner, not  
on every node

BROWSER

IPFS

DEVICE

## VALIDATION

the recipient verifies – no  
validator set in the path

CLIENT-SIDE

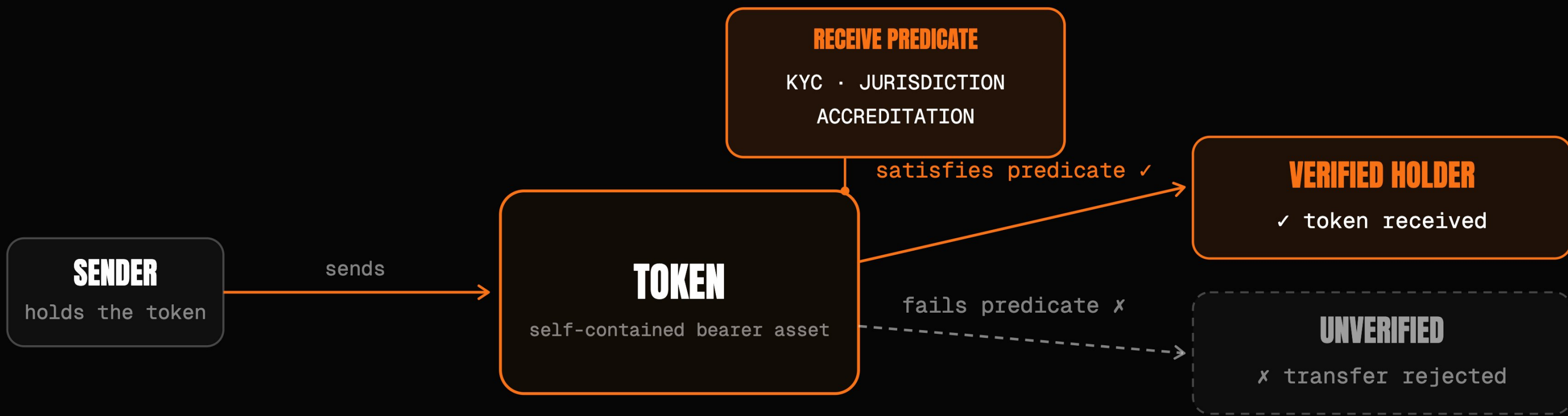
IN PARALLEL

🔗 See for yourself – [github.com/unicity-sphere/sphere-sdk](https://github.com/unicity-sphere/sphere-sdk)

# COMPLIANCE INTEGRATED INTO THE PROTOCOL

TOKENS CAN'T BE RECEIVED WITHOUT THE RIGHT CREDENTIALS.

ENTIRE HISTORY CAN BE VERIFIED COMPLIANT.



## RECEIVE PREDICATE

The token's own condition gates receipt  
– KYC, jurisdiction, accreditation.

## PROTOCOL-ENFORCED

Lives in the asset itself, not an app  
or custodian. No intermediary to  
bypass.

## ISSUER-DEFINED

The issuer sets who may hold it once;  
the rules travel with the token  
everywhere.

# TOTAL PRIVACY BY DEFAULT

THREE OBSERVERS. NONE CAN LEARN ANYTHING.

OBSERVER 01

## THE NETWORK

the Unicity service itself

**Sees only opaque commitments – never amounts, parties, or balances.**

A request is a hash of a state and a hash of transaction data. The nonce that chains one state to the next is hidden by a perfectly hiding commitment – so the service cannot link two consecutive transactions of the same token.

OBSERVER 02

## THE SENDER

the person who paid you

**Cannot watch when you later spend what they sent.**

Once a token reaches you, the previous owner has no way to compute the registry entry your next spend will create. The next state derives from a secret only you hold. Send it, then they're out.

OBSERVER 03

## ANYONE WHO KNOWS YOUR ADDRESS

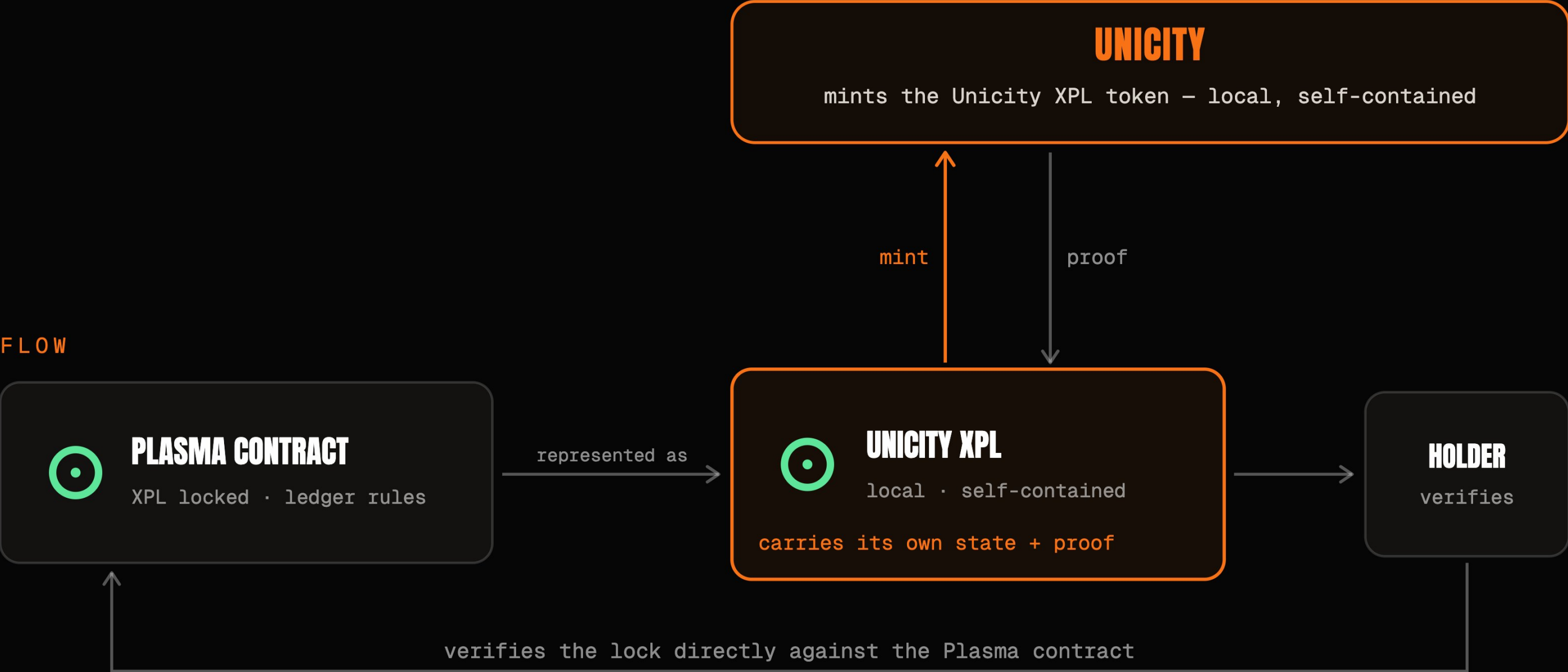
counterparties, partners, ex-employers

**Cannot link two payments to the same recipient – even when both went to you.**

You publish one persistent public key. Every sender derives a fresh, indistinguishable transaction key against it. No correlation, no UX overhead, no one-time-address management.

# BRIDGES ARE ELIMINATED: SOURCE TOKENS ARE VERIFIED DIRECTLY

A Unicity XPL token – verified straight from Plasma. No bridge to trust.

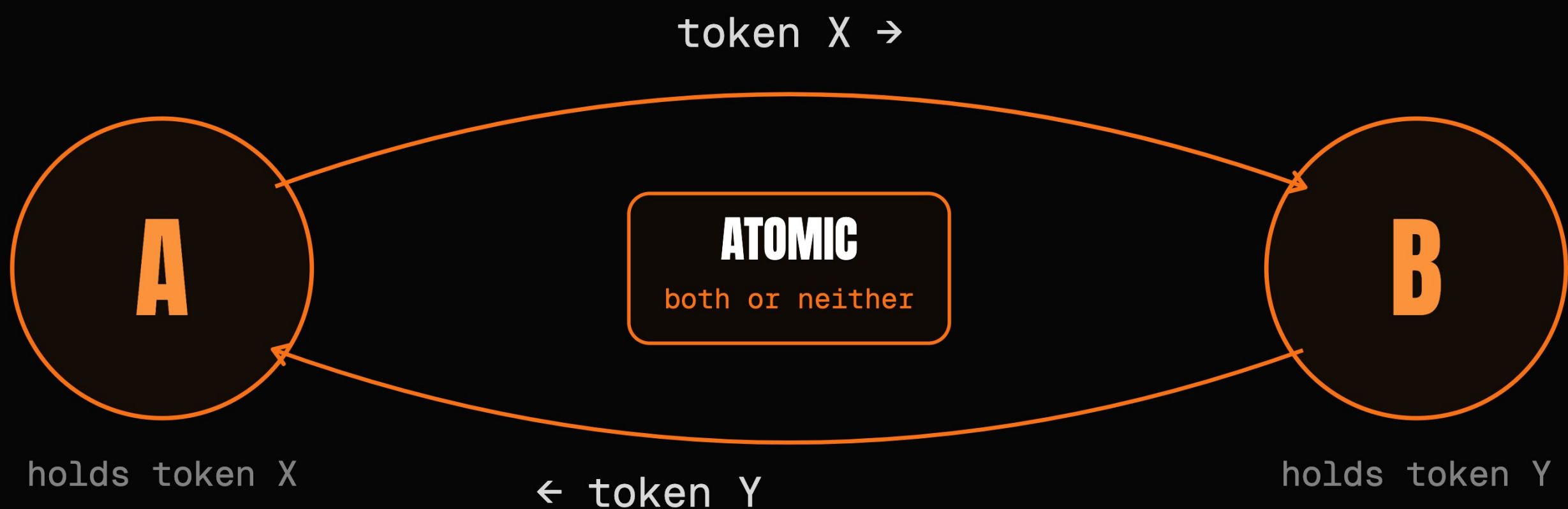


NO BRIDGE · NO CUSTODIAN · NOTHING TO HACK



# IN DEFI, THE ATOMIC OPERATION IS THE **SWAP**

Two assets change hands at once – **both legs, or neither.**



**Unicity has no shared ledger – so atomicity needs a new paradigm.**

*that's what predicates solve*

# THE UNICITY TRUSTLESS ATOMIC SWAP

Both parties commit independently – the swap completes, or everyone keeps their token.

## HASHED TIMELOCK (HTLC)

1 · A locks, reveals hash y

2 · B locks against same y

3 · A claims, revealing preimage x

4 · B must claim before timeout

### THE TIMING TRAP

B late on step 4 – a dropped connection – and A keeps both.

## UNICITY PREDICATE SWAP

### A COMMITS

locks their token

### B COMMITS

locks their token



### UNICITY SERVICE

the shared reference

### AUTOMATIC FAIRNESS

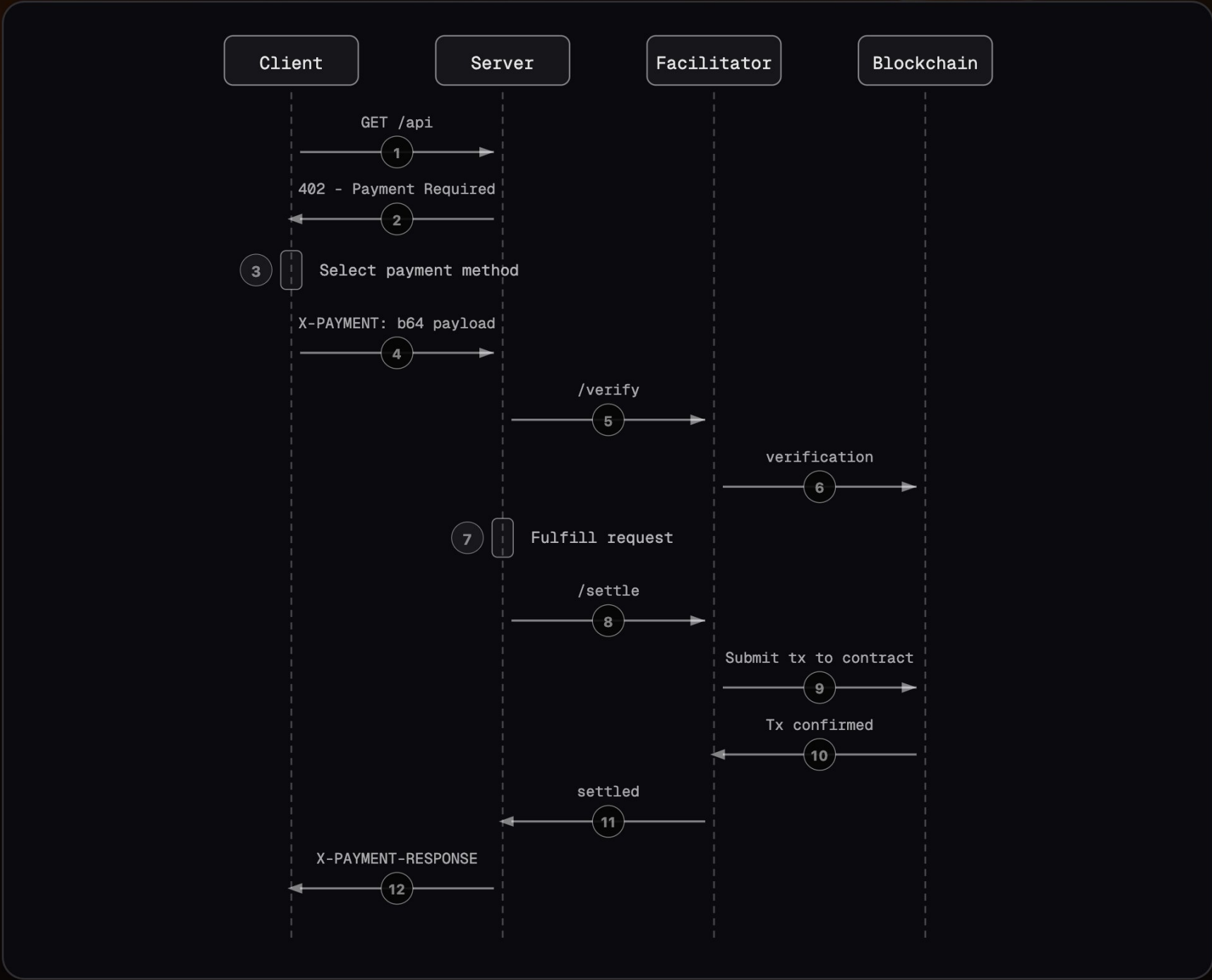
- ✓ both locked → swap completes
- ✓ either walked away → both refunded
- ✓ no deadlines, no chasing, done

# LIVE DEMO: X402 USDT AT 3M TPS WITH PRIVACY AND COMPLIANCE.

x402 Unicity implements and simplifies it – from 12 steps to 5.

[▶ Watch demo – USDT moving A2A at 3M TPS](#)

## TRADITIONAL BLOCKCHAIN X402 12 STEPS



## UNICITY FRICTIONLESS X402 5 STEPS





# CEX SPEED. DEX CUSTODY. PRIVATE AND COMPLIANT.

Binance has the first. Uniswap has the second. Neither has the third. Unicity has all three.

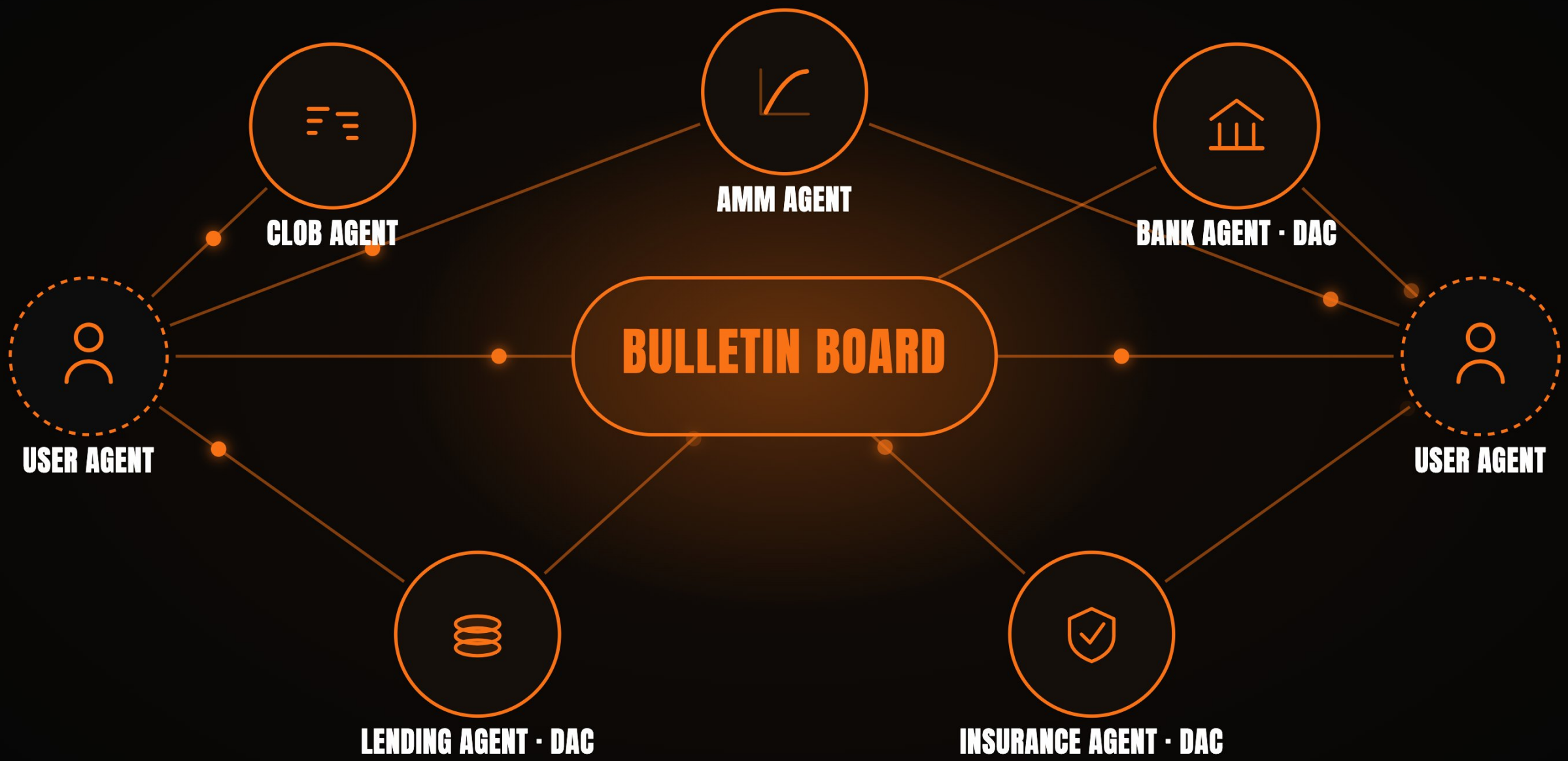
| PROPERTY       | BINANCE              | UNISWAP          | HYPERLIQUID         | UNICITY CLOB        |
|----------------|----------------------|------------------|---------------------|---------------------|
| Match speed    | ● <1 ms              | ○ 12 s block     | ● ~0.2 s            | ● ~0.2 s            |
| Gas per trade  | ● \$0                | ○ \$5-\$50       | ● \$0               | ● \$0               |
| Self-custody   | ○ Exchange-held      | ● Smart contract | ● Non-custodial     | ● You hold the keys |
| Privacy        | ○ Operator sees all  | ○ Fully public   | ○ Public positions  | ● Unlinkable        |
| MEV / sandwich | ● Internal front-run | ○ Public mempool | ● Validator-ordered | ● No mempool        |
| Settlement     | ○ Internal IOU       | ● On-chain       | ● Sub-second        | ● Sub-second        |
| Decentralized  | ○ Centralized        | ● Permissionless | ○ Admin keys        | ● Permissionless    |



# AGENTIC DEFI. **AN ECONOMY OF MACHINES.**

Frictionless. Massive throughput. Private. Compliant.

A financial system built for machines.



# THE INEVITABLE UNBUNDLING OF BLOCKCHAIN.

Each generation of consensus removed work from the network. We took the unbundling to its logical conclusion.

|                                                                                                                                                             |                                                                                                                                    |                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>2009</div> <div>BITCOIN</div> <div>Correctness + global ordering.</div> <div>Every node certifies every transaction. Every node agrees on order.</div> | <div>2023</div> <div>SUI / FASTPAY</div> <div>Correctness only.</div> <div>Validators certify correctness. Ordering removed.</div> | <div>2026</div> <div>UNICITY</div> <div>Uniqueness only.</div> <div>The network attests one thing: has this token been spent? Correctness moves to the edge.</div> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# DON'T TAKE OUR WORD FOR IT — HERE'S THE MATH

Three formal papers (LLM Friendly). Every claim is proven and public.

## SYSTEMS ENGINEERING

### Unicity Infrastructure: the Aggregation Layer

Risto Laanoja<sup>1</sup>

<sup>1</sup>Unicity Labs

May 22, 2026

#### Abstract

Unicity is a novel blockchain protocol with the ambitious goal of enabling token transactions to occur off-chain and, when necessary, offline. This promise requires supporting infrastructure to guarantee that there are no parallel states of assets, or more specifically, that there is no double-spending, a property we term the *unity*. It turns out that the lack of globally shared state and ordering reduces the blockchain overhead considerably. In designing the infrastructure, no compromises were made regarding its trust assumptions. This paper details the design of the Aggregation Layer, the component responsible for producing Proofs of Inclusion and Non-inclusion to the users. We analyze its design for efficiency and evaluate the robustness of its trust and security model. We then identify a single critical property that the Consensus Layer must verify in each round—*pre-state preservation*—and observe that the remaining desirable properties of the trustless Key-Value store (arbitrary tree shape, batch incorporation, no phantom inserts, completeness) are not required by the Unicity Aggregator's security model. This optimal statement is implemented [9] as Algebraic Intermediate Representation (AIR) circuit on top of the Plonky3 [34] STARK toolkit. The implementation sustains a proving throughput in excess of 10,000 signatures per second on a single consumer-class CPU, with a succinct proof and time of callstacks verification time, and no trusted setup.

#### 1 Motivation

The foundational principle of the Unicity Network [18] is to minimize the volume of on-chain data. This is based on the observation that shared ("on-chain") state is unavoidable only to prevent double-spending.<sup>1</sup> The core tenets of Unicity also include minimizing trust requirements, enhancing user privacy, and providing linear scale.

In a hierarchical trustless system, the principle is that the base layer (e.g., L1 blockchain) provides decentralization, while the layers below it (e.g., rollups) prevent cryptographic proofs of the correctness of their operation. In scaling Unicity, we have designed efficient data structures to prove the correctness of operation of Aggregation Layers to the Consensus Layer. Based on cryptographic hashes alone, the consistency proof grows linearly with respect to the number of user transactions. This requires a hard limit of approx. 10,000 transactions per second (tx/s), beyond which the networking bandwidth of the Consensus Layer becomes the bottleneck.

To scale further, we use cryptographic zero-knowledge proofs (ZKPs) to compress the size of the consistency proofs. As an application of ZKPs, this one-use is fundamentally more efficient than using ZKPs to process the transaction data itself, as is done in many privacy coins and ZK-rollups: the statement

<sup>1</sup> Assuming no externally controlled, non-transparent side channels such as trusted hardware modules (Trusted Execution Environments (TEEs)), and that anyone can be a recipient

off-chain & offline tokens  
ZK proofs · no trusted setup

 [github.com/unicitynetwork](https://github.com/unicitynetwork)

## SECURITY PROOF

### The Unicity Execution Layer

Ahto Baidas

Ahto Triin

Vladimir Rogojin

Risto Laanoja

November 2025

#### Abstract

This paper introduces the Unicity Execution Layer, a modular component of the Unicity framework enabling secure off-chain transactions while maintaining trustless double-spending prevention. We present a formal security model where token ownership is represented by public keys and transfers require digital signatures. We prove three fundamental security properties: (1) no double-spending—each token state can be spent at most once, (2) no blocking—only the legitimate owner can prevent a token from being spent, and (3) service-side privacy—the Unicity Service cannot link transactions with the same token. The service-side privacy is addressed by introducing generalized multi-policyable signatures wherein that allow one secret to generate multiple verifiable public keys, and interactive and non-interactive concrete instantiations, enabling private transaction with stable public identity with minimal key management overhead.

#### 1 Introduction

Blockchain technology has revolutionized digital asset management by enabling trustless peer-to-peer transactions without relying on centralized authorities. However, traditional blockchain architectures face fundamental scalability limitations that hinder their adoption for high-throughput applications. The core bottleneck stems from the fact that the "security" depends on the number of participating validators, which all have to participate in consensus on ordering, re-associate transactions, and store every previous block.

This paper introduces Unicity, a novel blockchain infrastructure designed to enable secure off-chain transactions while maintaining the trustless guarantees of traditional blockchains. The key insight underlying Unicity is that the vast majority of blockchain operations—transaction execution, smart contract processing, and state transitions—can be moved off-chain, leaving only the essential double-spending prevention mechanism on-chain. This also simplifies on-chain operations, making efficient and self-authenticating implementations possible.

By minimizing the data that must be processed by the consensus layer, Unicity achieves linear scalability while preserving the security properties that make blockchains trustworthy. The system consists of three hierarchical layers: the

1

no double-spend · no blocking  
service & user-side privacy

 [github.com/unicitynetwork](https://github.com/unicitynetwork)

## PREDICATES & SWAPS

### Unicity: Predicates and Atomic Swaps

Ahto Baidas

Ahto Triin

Vladimir Rogojin

Risto Laanoja

November 2025

#### Abstract

We generalize Unicity token ownership to programmable spending conditions called *predicates*, enabling smart-contract like functionality executed off-chain directly by moving parties rather than by consensus participants. We prove that the security properties of the Unicity execution layer are preserved under reductions to predicate validity undeniability. To demonstrate the utility of the model, we show how to implement trustless atomic swaps by using predicates.

#### 1 Introduction

Predicates generalize the concepts of token ownership and transfer in the Unicity infrastructure, which in the paper [3] were defined via digital signatures as follows:

- *Owner* – a (legal/physical) person who controls the private key of a digital signature scheme
- *Ownership condition* – the public key  $pk$  that corresponds to the private key of the owner
- *Transfer* – the owner presents a digital signature  $\sigma$  on  $m = H(h_{tx}, h_{tx})$  (where  $h_{tx}, h_{tx}$  are the state hash and the transaction hash, respectively) such that  $V(pk, m, \sigma) = 1$ .

In this paper, we present the following generalized concepts:

- *Owner* – an abstract group of (legal/physical) persons that together control the information necessary to create the next transaction with the token. The information may include private keys.
- *Ownership condition* – a logical condition (predicate)  $\psi$ .
- *Transfer* – the owner (as a group) presents a bit-string  $u$  such that the condition  $\psi(r, m, u) = 1$  holds, where  $m = H(h_{tx}, h_{tx})$  and  $r$  is the system time (an integer defined by the Unicity service). This clause that predicates may also put restrictions on transaction execution time.

programmable predicates  
trustless atomic swaps

 [github.com/unicitynetwork](https://github.com/unicitynetwork)



# WE BUILT THIS BEFORE.

Sovereign-grade cryptographic infrastructure, in production since 2012.

## THE TEAM

Built Guardtime / KSI – sovereign-grade verification with the Estonian Government, Lockheed Martin, Boeing and NATO. In production since 2012, eIDAS-grade. KSI held 300,000 tx/sec in Eesti Pank 2021 – the lineage, not a live Unicity number.

## THE PROOF

Three mathematical papers prove privacy and no-double-spend. Drop them into any model and they check. All public on [github.com/unicitynetwork](https://github.com/unicitynetwork).

## THE ALIGNMENT

Identity, speed, and peer-to-peer settlement, bound into one instrument – the fair-access infrastructure Hard Yaka has backed across a generation of payments companies.



# THE ASK: \$5M.

A seed to ship a compliant dollar that settles the moment the credential checks out.

## **\$5M SEED** - \$25M CAP / \$50M TOKEN FDV - SAFE + TOKEN WARRANT

First build: a USBC-style dollar that settles peer-to-peer the instant the Receive Predicate is satisfied – binding **identity**, **speed**, and **settlement** into one instrument.

Institutions rely on walled gardens for compliance. **Unicity embeds that compliance directly into the dollar.**